

# Oderdene Chuluunsukh

<https://www.linkedin.com/in/oderdene-chuluunsukh/>

[oderdene.galmaa@gmail.com](mailto:oderdene.galmaa@gmail.com)

Dallas, TX, US 75225

+1 (641) 233-9317

## Summary

Dependable Software Engineer with 9+ years driving success across the full SDLC, focused on backend and cloud-native development. Proficient in Python, Java and Javascript, adept at architecting and deploying scalable APIs, integrating production-ready ML-powered services, and building dynamic React.js, Bootstrap interfaces. Proven cross-functional collaborator leveraging AWS and Docker to deliver secure, performant, and reliable platform solutions.

## Skills

- Python, Java, JavaScript, TypeScript
- React, Next.js, Bootstrap
- FastAPI, Flask, Django, Spring Boot
- MySQL, PostgreSQL, MSSQL, AWS RDS
- AWS (EC2, ECS, SNS, SQS, Lambda, S3, CDN)
- Docker, Kubernetes, CI/CD, Github Actions, Google Cloud
- Nginx, Huey, Prometheus, Grafana, OpenTelemetry
- Pandas, NumPy, scikit-learn, Matplotlib, BeautifulSoup
- pytest, Jest, Selenium
- Jira, Trello, Confluence
- Data Structures and Algorithms, OOP, SOLID

## Professional Experience

### Senior Software Engineer | Nasha Tech LLC

09/2021 – 07/2024, Ulaanbaatar, Mongolia

- Engineered a high-performance microservices architecture using Python FastAPI and Docker, achieving sub-100ms API latency and seamless integration for cross-platform clients.
- Deployed and managed microservices using Kubernetes, reducing deployment downtime and optimizing resource utilization.
- Built a resilient media processing service using AWS S3, reducing large file upload failures by 30% and ensuring secure storage for sensitive user documents and training videos.
- Architected a normalized PostgreSQL schema and optimized complex queries, improving data retrieval speeds by 20%.
- Developed a dynamic QR code verification system, cutting manual health compliance checks at restaurants by 40%.
- Led the full-stack development of the pet tech platform using React and React Native, ensuring seamless data synchronization with the backend services.
- Engineered a closed Social Network Service for medical researchers with Next.js and Firestore, ensuring secure, compliant collaboration and reducing project communication overhead.
- Designed and deployed an AI-driven scheduling module powered by the OpenAI API, reducing doctors' night shift conflicts.
- Fine-tuned a YOLO object detection model to identify a dog's muzzle from images and infer its health status, achieving a 61% accuracy rate as part of a proof-of-concept module.

### Senior Software Engineer | Baatar Khangamj LLC

08/2020 – 08/2021, Ulaanbaatar, Mongolia

- RESTful APIs with Python Flask and Docker to crawl real-time COVID-19 data from government websites, utilizing multi-threading and fault-tolerant logic to ensure reliability, enhancing data accessibility.
- Optimized data storage and retrieval with MySQL using advanced indexing, normalization, and query optimization strategies, significantly improving read/write performance and data integrity.
- Used Postman and pytest for comprehensive API testing, while Git provided robust version control and seamless collaborative development, ensuring high-quality software delivery.

### Software Developer | Dentsu Data Artist Mongol LLC

04/2018 – 08/2020, Ulaanbaatar, Mongolia

- Applied Tesseract OCR with EAST (TensorFlow) to detect text in images, achieving 60% accuracy for English and ~30% for Japanese depending on font type, validating feasibility for Japanese OCR use cases.
- Processed uploaded videos using OpenCV pipelines to extract frames and convert them into analyzable images, enabling scalable video-to-image content analysis for Minimum Viable Product development.
- Web scraping solutions using Python and BeautifulSoup to aggregate data from diverse sources (news, Olympic sites, Wikipedia, and SNS), providing comprehensive datasets for analysis.
- Automated the data ingestion pipeline using AWS Lambda and API Gateway, enabling continuous, timely updates from target websites and effectively reducing manual data collection efforts by over 30%.
- Designed and deployed robust data processing and storage architecture on AWS with Flask and MySQL, automating the cleansing and organization of collected data to support keyword relationship analysis for marketing strategies.
- Analyzed mobile phone location data in BigQuery and Python/Pandas to infer demographic attributes (age, gender, occupation); identified limitations of antenna-based data, improving data quality assessment practices.
- Developed and fine-tuned a Japanese BERT model on a labeled dataset, achieving 85% classification accuracy for unseen text, significantly improving language-specific NLP capabilities.

- Maintained and enhanced the university’s core student information system used by over 30,000 active users.
- Contributed to system modernization by migrating the frontend from legacy Spring with jQuery and Bootstrap, improving UI responsiveness and maintainability.
- Refactored and optimized complex MSSQL stored procedures and triggers to improve system performance and data accuracy.
- Developed and maintained key modules—including Faculty, Student Requests, and Transfer & Leave—using Spring Boot to support academic operations and communication.

Key Projects

- *Pet Tech Platform*: Cross-platform solution (React, FastAPI, PostgreSQL, AWS) for pet vaccination records, trainer bookings, restaurant discovery, and QR-enabled compliance verification.
- *Medical Knowledge Platform*: Developed secure HealthTech/EdTech platform for surgeons to exchange expertise and training resources in a trusted digital environment.
- *2020 Olympic Data Analytics*: Automated web scraping and across Olympic websites and Social Network Service, reducing manual data collection and powering marketing keyword analysis.
- *Video-meta*: (Tesseract, EAST) and OpenCV pipelines for video frame analysis, achieving up to 60% text detection accuracy in English.
- *School Management System*: Maintained and modernized the university's core student information system, migrating its legacy frontend and optimizing complex MSSQL stored procedures to enhance overall system performance.

Education

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|-----------------------------------------------------------|-----------------------|
| MS Computer Science   Maharishi International University  | Fairfield, IA, US     |
| BS Computer Science   The National University of Mongolia | Ulaanbaatar, Mongolia |